GMAT Word-Problem Translation Workbook

The 4-Step Mathematical Translation Method, Clue-to-Operator Dictionary & Worked Problem Archetypes

1. Verbal Clues to Mathematical Operators Dictionary

Translate complex English sentences into algebraic equations with zero ambiguity using this standardized dictionary.

English Word / Phrase	Mathematical Operator / Symbol	Algebraic Setup Example	Common Translation Trap
"is, was, will be, represents, equals"	Equal sign (=)	"John is 5 years older than Mary" -> J = M + 5	Writing J + 5 = M (inverting the subject relationship)
"more than, increased by, exceeds by, sum of"	Addition (+)	"x exceeds y by 12" -> x - y = 12 or x = y + 12	Confusing "exceeds by" with multiplication
"less than, decreased by, difference between"	Subtraction (-)	"6 less than three times p" -> 3p - 6	Writing 6 - 3p (order matters in subtraction!)
"of, times, product of, fraction of, percent of"	Multiplication (*)	"35% of total revenue R" -> 0.35 * R	Forgetting to divide percentage by 100
"per, out of, ratio of x to y, divided by"	Division (/)	"Ratio of boys to girls is 4 to 5" -> B / G = 4 / 5	Flipping numerator and denominator
"at least, no less than, minimum"	Greater than or equal to (>=)	"Revenue is at least $50,000" -> R >= 50000	Using strict greater than (>) instead of >=
"at most, no more than, maximum"	Less than or equal to (<=)	"Capacity is at most 120 units" -> C <= 120	Using strict less than (<) instead of <=

Golden Rule: Always assign clear variable names (e.g. J for John, M for Mary) instead of generic x and y.

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2. The 4-Step Systematic Word Problem Translation Method

Follow this exact sequence on every word problem to prevent mental overload.

Step Number	Action Item	Execution Method	Verification Checkpoint
Step 1: Read for Context	Identify the story scenario (speed, profit, mixtures, age)	Do not write equations yet; visualize the real-world scenario	What are the physical entities involved?
Step 2: Define Variables	Write down what the question explicitly asks you to solve for	State: "Let T = total time in hours"	Are units consistent (minutes vs hours, dollars vs cents)?
Step 3: Translate Relationships	Convert each sentence into an algebraic equation using the dictionary	Write equations on scratchpad line by line	Do you have as many distinct equations as unknowns?
Step 4: Solve & Sanity Check	Perform algebraic substitution or elimination	Solve for the requested variable, not just x	Does the answer make physical sense (positive age, speed > 0)?

Pacing Target: Complete Steps 1-3 within 45 seconds; spend remainder of time on clean algebraic resolution.

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3. High-Frequency Worked Archetypes & Translation Traps

Key formulas and structural setups for the top 3 GMAT word problem categories.

- Work/Rate Formula: Work = Rate * Time. For combined workers: 1/T_combined = 1/T_a + 1/T_b. Always solve using rates (work per hour), never by averaging times directly.
- Distance/Speed Formula: Distance = Speed * Time. Average Speed = Total Distance / Total Time. (Never take the arithmetic mean of two speeds!).
- Overlapping Sets Formula: Total = Group A + Group B - Both + Neither. For 3 groups: Total = A + B + C - (sum of 2-group overlaps) + (all 3) + Neither.
- Mixture Alligation Shortcut: Ratio of Quantities = (Price of Dearer - Mean Price) / (Mean Price - Price of Cheaper).